



Advanced Training for Oil Tanker Cargo Operations (TASCO)

201 Practice Questions and Answers for STCW Exams

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Practice Set 1

Q1. What is the main requirement of the OPA 90?

- A) New tankers and tank barges should have double hulls
- B) New tankers and tank barges should not ply in US waters
- C) New tankers and tank barges with single hulls can ply in us ports

Correct Answer: A (New tankers and tank barges should have double hulls)

Explanation: The Oil Pollution Act of 1990 (OPA 90) was enacted in the United States following the Exxon Valdez incident. Its primary requirement is the mandatory fitting of double hulls on all new tankers and tank barges to prevent oil spills during groundings or collisions.

Q2. Volume 2 of IMDG code comprises one part which is

- A) Part 1 - General provisions, definitions and training
- B) Part 2 - Classification
- C) Part 3 - The Dangerous Goods List (DGL)
- D) Part 4 - Packing and tank provisions

Correct Answer: C (Part 3 - The Dangerous Goods List (DGL))

Explanation: Volume 2 of the IMDG Code is structured to provide technical guidance, specifically containing Part 3, which is the Dangerous Goods List (DGL). This list is essential for identifying the specific shipping requirements and classification of hazardous materials.

Q3. What is the maximum limit of sulphur content in bunker fuel oil?

- A) 4.5% m/m
- B) 1.5% m/m
- C) 3.0% m/m

Correct Answer: A (4.5% m/m)

Explanation: MARPOL Annex VI sets strict limits on the sulphur content of bunker fuel to reduce SOx emissions. The standard global limit, as established in the regulation, is 4.5% m/m, though regional Emission Control Areas (ECAs) now have significantly stricter requirements.

Q4. What is the minimum requirement for carrying garbage management plan onboard a ship?

- A) Any ship above 400 GT or any ship carrying 15 persons or more
- B) Any ship which is 12 metres or more in length
- C) All ship, Irrespective of size and length

Correct Answer: C (All ship, Irrespective of size and length)

Explanation: According to MARPOL Annex V, all ships, regardless of size and length, are required to have a Garbage Management



~~Q5. Petroleum liquids that have closed-cup flash points below 140 Deg F (60 Deg C) are considered as~~

- A) Non-volatile Liquids
- B) Flammable Liquids
- C) Volatile Liquids
- D) Combustible Liquids

Correct Answer: C (Volatile Liquids)

Explanation: Under standard classification, petroleum liquids with a closed-cup flash point below 60 degreesC are categorized as 'volatile' or 'flammable' because they can easily produce vapours that form ignitable mixtures with air at ambient temperatures.

Q6. What does SECA stand for?

- A) Sox Elimination Controlled Area
- B) Sox Emission Controlled Area
- C) Sox Escalation Controlled Area

Correct Answer: B (Sox Emission Controlled Area)

Explanation: SECA stands for SOx Emission Control Area. Under MARPOL Annex VI, these are designated sea areas where stricter controls on sulfur emissions from ships are enforced to reduce air pollution and protect human health and the environment.

Q7. What are the three broad classification of cargoes transported on tankers?

- A) Liquefied Natural gas, Liquefied Petroleum gas, Liquefied Nitrogen
- B) Container, Dry Bulk and Refrigerated cargo
- C) Petroleum liquids, chemical liquids, and special liquids
- D) General cargo, Dry Break Bulk and Frozen cargo

Correct Answer: C (Petroleum liquids, chemical liquids, and special liquids)

Explanation: Tanker cargoes are broadly classified into petroleum liquids (crude oil and products), chemical liquids (noxious liquid substances), and special liquids (like vegetable/animal oils and molasses). This classification helps determine carriage requirements, hazard profiles, and equipment standards under IBC and IGC codes.

Q8. What is the closed-cup flash point of non-volatile petroleum liquids?

- A) below 140 Deg F (60 Deg C)
- B) 113 Deg F (45 Deg C) and above
- C) below 113 Deg F (45 Deg C)
- D) 140 Deg F (60 Deg C) and above

Correct Answer: D (140 Deg F (60 Deg C) and above)

Explanation: Non-volatile petroleum liquids are defined by their higher flash points. According to international safety standards for tanker operations, a liquid with a closed-cup flash point of 60 degreesC (140 degreesF) or above is considered non-volatile, reducing the risk of vapor accumulation at ambient temperatures.

Q9. What is the Reid vapor pressure of flammable liquids of natural gasoline & Naphtha which has open cup flash point of 80 Deg F (26.7 Deg C)?

- A) More than 8.5 but less than 14 psi
- B) 8.5 psi and below
- C) 14 psi and above
- D) More than 24.6 psi

Correct Answer: A (More than 8.5 but less than 14 psi)

Q10. What is the requirement for ships to have SOPEP onboard ships?

- A) SOPEP Should be carried on board tankers of 150 GT or more and on other ships of 400 GT or more
- B) SOPEP should be carried onboard tankers of 150 GT or more and on all other ships irrespective of size
- C) SOPEP should be carried only onboard tankers of 400 GT or more

Correct Answer: A (SOPEP Should be carried on board tankers of 150 GT or more and on other ships of 400 GT or more)

Explanation: Under MARPOL Annex I, Regulation 37, every oil tanker of 150 GT or more and every ship of 400 GT or more is required to carry an approved Shipboard Oil Pollution Emergency Plan (SOPEP) to facilitate effective response to oil spills.

Q11. What is the meaning of a Category 2 ship?

- A) Oil tanker of 5000 tones deadweight and above
- B) Tanker of 20,000 deadweight and above without SBTs
- C) Tanker of 20,000 deadweight and above with SBTs

Correct Answer: B (Tanker of 20,000 deadweight and above without SBTs)

Explanation: Under MARPOL Annex I, Category 2 ships refer to oil tankers of 20,000 tonnes deadweight and above that are not fitted with Segregated Ballast Tanks (SBT) in accordance with the regulation's phase-out requirements for single-hull tankers.

Q12. What type of ships have to carry the SMPEP for NLS manual onboard their ships?

- A) Ship of 400 GT or more carrying 15 persons or more
- B) Ships of 150 GT or more carrying oil cargoes
- C) Ships of 150 Gt or more carrying noxious liquid substances

Correct Answer: C (Ships of 150 Gt or more carrying noxious liquid substances)

Explanation: Under MARPOL Annex II, ships of 150 GT or more that are certified to carry noxious liquid substances in bulk are required to have a Shipboard Marine Pollution Emergency Plan (SMPEP) for NLS on board.

Q13. Tankers carry palm oil, molasses and tallow as special liquids cargoes in bulk, which are grouped as

- A) Animal/Vegetable oils
- B) Miscellaneous Liquids
- C) Non-volatile Liquids
- D) Other chemical Liquids

Correct Answer: A (Animal/Vegetable oils)

Explanation: Palm oil, molasses, and tallow are classified as Animal and Vegetable oils. These are distinct from mineral oils and fall under specific requirements for carriage and pollution prevention on specialized tankers.

Q14. Under which Annex is the Garbage record book required?

- A) Annex II
- B) Annex V
- C) Annex VI

Correct Answer: B (Annex V)

Explanation: MARPOL Annex V deals with the prevention of pollution by garbage from ships. This regulation mandates the maintenance of a Garbage Record Book to track the discharge or disposal of all garbage generated onboard.



- A) Chemical Liquid
- B) Special Liquids
- C) Non-volatile Liquids
- D) Non-flammable Liquids

Correct Answer: B (Special Liquids)

Explanation: Animal and Vegetable oils, while distinct from petroleum hydrocarbons, are categorized as special liquids due to their persistent nature and specific requirements for carriage and potential pollution mitigation as defined in MARPOL and IBC Code guidelines.

Q16. Which part of the oil record book deals with machinery space operations?

- A) Part I
- B) Part II
- C) Part III

Correct Answer: A (Part I)

Explanation: The Oil Record Book is divided into two parts: Part I (Machinery space operations) for all ships, and Part II (Cargo/ballast operations) for oil tankers. All entries regarding oil transfers, fuel bunkering, and bilge water discharge are strictly documented here.

Q17. Structure of MARPOL Convention layouts are as follows: find the correct one.

- A) Convention, Protocol I, Protocol II, Protocol 1978, Protocol 1997
- B) Convention, Protocol 1978, Protocol 1997, Protocol I, Protocol II
- C) Protocol 1978, Protocol 1997, Protocol I, Protocol II, Convention
- D) Protocol 1997, Protocol 1978, Protocol I, Protocol II, Convention

Correct Answer: B (Convention, Protocol 1978, Protocol 1997, Protocol I, Protocol II)

Explanation: The formal structure of the MARPOL Convention includes the main Convention, Protocol I (Reports on Incidents), Protocol II (Arbitration), and subsequent major amendments including the 1978 Protocol (MARPOL 73/78) and the 1997 Protocol (Annex VI).

Q18. The IMDG Code became mandatory on

- A) 1st of January 2000
- B) 1st of January 2002
- C) 1st of January 2004
- D) 1st of January 2006

Correct Answer: C (1st of January 2004)

Explanation: The International Maritime Dangerous Goods (IMDG) Code was adopted by the IMO and became mandatory under SOLAS Chapter VII on 1st January 2004 to ensure the safe transport of dangerous goods by sea.

Q19. How should CAS be aligned with other ship's surveys?

- A) Alignment with the renewal survey and no inclusion in the drydock survey
- B) Alignment with the intermediate or special survey and no inclusion in the drydock survey
- C) Independent of the intermediate survey and drydock survey

Correct Answer: B (Alignment with the intermediate or special survey and no inclusion in the drydock survey)

Explanation: The Condition Assessment Scheme (CAS) for tankers is a mandatory survey requirement for certain ships. It is aligned with the ship's enhanced survey program, typically during the intermediate or special renewal survey, to ensure structural integrity.



Q20. Liquids that have an open-cup flash point at or below 80 Deg F (26.7 Deg C) are classified as

- A) Non-volatile Liquids
- B) Flammable Liquids
- C) Volatile Liquids
- D) Combustible Liquids

Correct Answer: B (Flammable Liquids)

Explanation: In accordance with safety standards, liquids with an open-cup flash point at or below 80 degreesF (26.7 degreesC) are classified as 'flammable liquids' due to their ability to form explosive gas mixtures at ambient temperatures.

Q21. Which Annex deals with the prevention of pollution by sewage from ships?

- A) Annex VI
- B) Annex III
- C) Annex IV

Correct Answer: C (Annex IV)

Explanation: MARPOL Annex IV provides international requirements for the prevention of pollution by sewage from ships, covering discharge rates, treatment plants, and equipment standards for various ship types.

Q22. What type of pollution control is addressed by Annex II of MARPOL?

- A) Pollution by oil
- B) Pollution by noxious liquid substances
- C) Pollution by garbage

Correct Answer: B (Pollution by noxious liquid substances)

Explanation: MARPOL Annex II specifically addresses the control of pollution by Noxious Liquid Substances (NLS) carried in bulk, establishing categorization systems (X, Y, Z, OS) for chemicals to mitigate environmental damage.

Q23. What was the last date by which use of hydrochloroflourocarbons were prohibited?

- A) 2015
- B) 2010
- C) 1 Jan 2020

Correct Answer: C (1 Jan 2020)

Explanation: Under MARPOL Annex VI regulations concerning ozone-depleting substances, the use and installation of equipment containing hydrochlorofluorocarbons (HCFCs) were strictly prohibited globally from 1st January 2020.

Q24. When did Annex VI of Marpol enter into force?

- A) 1 July 1992
- B) 19 May 2006
- C) 31 December 1988

Correct Answer: B (19 May 2006)

Explanation: MARPOL Annex VI, which deals with the prevention of air pollution from ships by regulating sulfur oxides, nitrogen oxides, and ozone-depleting substances, officially entered into force on 19 May 2006.



- A) Above 80 Deg F (26.7 Deg C)
- B) Above 60 Deg F (15.6 Deg C)
- C) Below 80 Deg F (26.7 Deg C)
- D) Below 60 Deg F (15.6 Deg C)

Correct Answer: A (Above 80 Deg F (26.7 Deg C))

Explanation: Combustible liquids are those with a flash point higher than 80 degreesF (26.7 degreesC). Because they do not readily vaporize at ambient conditions, they are considered less immediately hazardous than flammable liquids but still require safety precautions.

Q26. ____ is a lowest temperature at which a substance catches fire.

- A) Critical temperature
- B) Ignition temperature
- C) Eutectic temperature

Correct Answer: B (Ignition temperature)

Explanation: Ignition temperature is the minimum temperature to which a substance must be heated so that it will self-sustain combustion without the need for an external spark or flame.

Q27. What is an inert condition?

- A) When the oxygen has been eliminated from a tank
- B) When the oxygen is 10% by volume in a tank
- C) A condition in which the oxygen content throughout the atmosphere of a tank has been reduced to 8% or less by volume by the addition of inert gas.
- D) When the oxygen is 21% by volume in a tank

Correct Answer: C (A condition in which the oxygen content throughout the atmosphere of a tank has been reduced to 8% or less by volume by the addition of inert gas.)

Explanation: An inert condition is defined as an atmosphere where the oxygen content is reduced to 8% or less by volume, usually by injecting flue gas or nitrogen, making the tank atmosphere non-combustible.

Q28. What factor mainly affects Vapour cloud dilution rate?

- A) Quantity of Spill
- B) Weather conditions
- C) Thermal inversion
- D) Diameter of the vapour cloud

Correct Answer: B (Weather conditions)

Explanation: Weather conditions, specifically wind speed and direction, are the primary drivers for vapor cloud dispersion and dilution. They dictate how quickly a released gas plume is mixed with fresh air, reducing its concentration below the LFL.

Q29. The substance involved in combustion is called

- A) Combustible
- B) Non-combustible
- C) Inflammable substances

Correct Answer: A (Combustible)

Explanation: The substance that undergoes the chemical reaction of oxidation during fire is technically referred to as the combustible material or fuel.



- A) Carrying out work in a cold atmosphere
- B) Work that cannot create a source of ignition
- C) Carrying out work on reefer containers
- D) Work that can be done only in cold temperatures

Correct Answer: B (Work that cannot create a source of ignition)

Explanation: Cold work refers to maintenance activities that do not involve heat, sparks, or potential sources of ignition, such as painting or manual cleaning, thereby not requiring a 'Hot Work' permit.

Q31. _____ burns automatically at room temperature.

- A) Phosphorus
- B) Sodium
- C) Iron
- D) Nitrogen

Correct Answer: A (Phosphorus)

Explanation: Phosphorus is a highly reactive chemical element that can spontaneously ignite when exposed to air at ambient room temperatures, requiring specialized storage and handling procedures.

Q32. Which type of anodes must not be fitted in tanks where flammable gases can be present?

- A) Magnesium anodes
- B) Aluminium anodes
- C) Zinc anodes

Correct Answer: A (Magnesium anodes)

Explanation: Magnesium anodes are prohibited in cargo tanks where flammable gases may be present because they can generate incendiary sparks upon impact, which could lead to a catastrophic explosion.

Q33. What is meant by "intrinsically safe"?

- A) Equipment can be used in hazardous atmosphere
- B) An electrical circuit is intrinsically safe if any spark produced normally or accidentally is incapable of igniting a gas mixture
- C) There is no danger if sparks are produced in an unsafe atmosphere

Correct Answer: B (An electrical circuit is intrinsically safe if any spark produced normally or accidentally is incapable of igniting a gas mixture)

Explanation: Intrinsically safe equipment is designed so that the electrical and thermal energy generated by the device-even under fault conditions-is insufficient to ignite a specific hazardous gas mixture.

Q34. What is a hot work permit?

- A) Document used for carrying out repairs under supervision
- B) A document issued by a responsible person permitting specific hot work to be done during a particular time interval in a defined space
- C) Document used for work in a drydock
- D) Document valid for 7 days for carrying out tank cleaning

Correct Answer: B (A document issued by a responsible person permitting specific hot work to be done during a particular time interval in a defined space)

Explanation: A hot work permit is a critical safety document issued by the responsible officer that authorizes specific work involving heat or sparks in a designated space only after safety measures are verified.

Q35. What is bonding?

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- A) Connecting together of cargo pipe line flanges
- B) The connecting together of metal parts to ensure electrical continuity
- C) Making a ship to shore cable connection
- D) Connecting together flexible couplings on deck

Correct Answer: B (The connecting together of metal parts to ensure electrical continuity)

Explanation: Bonding is the electrical connection between metal parts of a system to ensure electrical continuity, which prevents the build-up of static electricity and potential spark discharges during cargo operations.

Q36. How Threshold Limit Values are specified?

- A) PPM (Parts per Million)
- B) Specific Numbers
- C) Volume of Gas in Air
- D) ppm/volume gas in Air

Correct Answer: A (PPM (Parts per Million))

Explanation: Threshold Limit Values (TLV) represent the concentration of substances to which personnel can be exposed daily without adverse effects; these are standardly expressed in Parts per Million (PPM).

Q37. What is pour point?

- A) The temperature at which oil gets frozen
- B) The lowest temperature at which a petroleum oil will remain fluid
- C) The temperature at which oil cannot be discharged
- D) The temperature at which oil needs to be heated

Correct Answer: B (The lowest temperature at which a petroleum oil will remain fluid)

Explanation: Pour point is the lowest temperature at which a liquid (oil) remains fluid enough to pour under specified test conditions, which is critical for determining pumpability in cold climates.

Q38. A liquid is considered to be flammable when the flash point is less than

- A) 50°F
- B) 60°F
- C) 70°F
- D) 80°F

Correct Answer: D (80°F)

Explanation: A liquid is defined as flammable in maritime safety standards if its closed-cup flash point is 60 degreesC or lower (often rounded to 140 degreesF), but 80 degreesF is the specific limit frequently cited for high-volatility classification.

Q39. What is Upper flammable limit?

- A) The limit for man entry of personnel into tanks
- B) The concentration of a hydrocarbon gas in air above which there is insufficient oxygen to support combustion
- C) The limit where there is insufficient inert gas to support combustion
- D) The limit where there is too much inert gas to support combustion

Correct Answer: B (The concentration of a hydrocarbon gas in air above which there is insufficient oxygen to support combustion)

Explanation: The Upper Flammable Limit (UFL) is the maximum concentration of a hydrocarbon gas in air above which the mixture is too 'rich' (insufficient oxygen) to support combustion.



- A) The concentration of a hydrocarbon gas in air below which there is insufficient hydrocarbon to support combustion
- B) The limit where there is insufficient oxygen to support combustion
- C) The limit where there is too much oxygen to support combustion
- D) The limit for man entry of personnel into tanks

Correct Answer: A (The concentration of a hydrocarbon gas in air below which there is insufficient hydrocarbon to support combustion)

Explanation: The Lower Flammable Limit (LFL) is the lowest concentration of a gas in air that can support combustion; below this limit, the mixture is too 'lean' (insufficient fuel) to ignite.

Q41. What is cathodic protection?

- A) Cathodes placed on the hull of a ship
- B) Anodes placed in the stern of a ship
- C) Protection of the hull of a ship
- D) The prevention of corrosion by electrochemical techniques

Correct Answer: D (The prevention of corrosion by electrochemical techniques)

Explanation: Cathodic protection is an electrochemical technique used to control the corrosion of a metal surface (like a ship's hull) by making that surface the cathode of an electrochemical cell using sacrificial anodes.

Q42. Where is smoking permitted on board ship while at a berth?

- A) Designated smoking room
- B) Anywhere on the ship
- C) In the engine room
- D) In the cargo control room

Correct Answer: A (Designated smoking room)

Explanation: Smoking is strictly prohibited on tankers except in specially designated, safe, and controlled areas, as any ignition source near cargo spaces poses a significant fire and explosion risk.

Q43. As a requirement of SOLAS, the number of sets of full protective clothing resistant to chemical attack that should be provided in addition to firefighters' outfits, is

- A) 2 sets
- B) 3 sets
- C) 4 sets
- D) 5 sets

Correct Answer: A (2 sets)

Explanation: SOLAS regulations require ships carrying dangerous chemical cargoes to have at least two sets of full protective clothing resistant to chemical attack, separate from standard firefighter outfits, to ensure safety during incidents.

Q44. What is a material safety data sheet (MSDS)?

- A) It is used for finding out the dangers of contact
- B) It is used for finding out the effects of inhalation
- C) A document identifying a substance and all its constituents
- D) A document giving the chemical name of the product

Correct Answer: C (A document identifying a substance and all its constituents)

Explanation: A Material Safety Data Sheet (MSDS) is an essential document provided by manufacturers that lists the properties, hazards, safe handling procedures, and emergency response information for a specific chemical substance.



- A) Hot work is carried out in high ambient temperatures
- B) Work involving sources of ignition or temperatures sufficiently high to cause the ignition of a flammable gas mixture.
- C) The temperature is very high where the work is done
- D) Work carried out using hot material

Correct Answer: B (Work involving sources of ignition or temperatures sufficiently high to cause the ignition of a flammable gas mixture.)

Explanation: Hot work is defined as any activity that can produce a source of ignition-such as welding, burning, or grinding-in areas where flammable vapors might be present.

Q46. What is volatile petroleum?

- A) Petroleum having a flashpoint above 60 Degrees C
- B) Petroleum having a flashpoint above 100 Degrees C
- C) Petroleum having a flashpoint below 100 Degrees C
- D) Petroleum having a flashpoint below 60 Degrees C

Correct Answer: D (Petroleum having a flashpoint below 60 Degrees C)

Explanation: Volatile petroleum is defined as any petroleum product with a flash point below 60 degreesC, meaning it can easily form flammable vapor mixtures at normal ambient temperatures.

Q47. What is flash point?

- A) The lowest temperature at which a liquid gives off sufficient gas to form a flammable gas mixture near the surface of the liquid.
- B) The temperature at which the gas is not flammable
- C) The temperature at which the liquid stops emitting gas
- D) The temperature at which an explosion is not possible

Correct Answer: A (The lowest temperature at which a liquid gives off sufficient gas to form a flammable gas mixture near the surface of the liquid.)

Explanation: Flash point is the lowest temperature at which a liquid emits sufficient vapor to form an ignitable mixture with air near the liquid's surface, as determined by standardized laboratory tests.

Q48. What is non-volatile petroleum?

- A) Petroleum having a flashpoint of 60 Degrees C or above
- B) Petroleum having a flashpoint of 60 Degrees C or below
- C) Petroleum having a flashpoint of 100 Degrees C or above
- D) Petroleum having a flashpoint of 100 Degrees C or below

Correct Answer: A (Petroleum having a flashpoint of 60 Degrees C or above)

Explanation: Non-volatile petroleum is defined as petroleum having a flash point of 60 degreesC (140 degreesF) or above, meaning it is less likely to form dangerous vapor concentrations under standard operational temperatures.

Q49. What is purging?

- A) The introduction of inert gas into a tank already in the inert condition to further reduce the existing oxygen content or hydrocarbon gas content to a level below which combustion cannot take place if air is introduced
- B) Introducing of oxygen into a tank to a high level
- C) Using blowers to push oxygen into a tank
- D) Using natural ventilation to gas free a tank

Correct Answer: A (The introduction of inert gas into a tank already in the inert condition to further reduce the existing oxygen content or hydrocarbon gas content to a level below which combustion cannot take place if air is introduced.)

Explanation: Purging is the process of displacing an existing tank atmosphere (usually hydrocarbon gas) with inert gas to further reduce the oxygen concentration, ensuring that any subsequent introduction of air will not create a flammable mixture.



A) Corona is a diffuse discharge from a single sharp conductor that slowly releases some of the available energy.

Generally, corona on its own is capable of igniting a gas.

B) Corona is a diffuse discharge from a single sharp conductor that slowly releases some of the available energy.

Generally, corona on its own is incapable of igniting a gas.

C) Corona is a diffuse discharge from a single sharp conductor that slowly releases some of the available energy.

Generally, corona on its own is incapable of igniting a gas if it is a lean mixture

Correct Answer: B (Corona is a diffuse discharge from a single sharp conductor that slowly releases some of the available energy)

Explanation: Corona is a discharge caused by ionization of air near sharp conductors; however, in standard tanker operations, a corona discharge alone does not possess sufficient energy to ignite a hydrocarbon gas-air mixture.

Practice Set 2

Q1. What is putrefaction?

A) Most animal and vegetable oils undergo decomposition over time in stored condition

B) Most animal and vegetable oils undergo decomposition over time in stored condition, when heated

C) Most animal and vegetable oils undergo decomposition over time in stored condition, when heated beyond a particular temperature which varies for each and every oil

Correct Answer: A (Most animal and vegetable oils undergo decomposition over time in stored condition)

Explanation: Putrefaction is the process of decomposition of organic matter, such as animal and vegetable oils, which can lead to the release of toxic or flammable gases when stored for long periods.

Q2. Fuel must be heated to its _____ before it starts burning.

A) ignition temperature

B) flash point

C) fire point

D) boiling point

Correct Answer: A (ignition temperature)

Explanation: Fuel must reach its ignition temperature through external heat transfer for a sustained combustion reaction to begin; this is the temperature where the chemical reaction becomes self-sustaining.

Q3. What is Nitrogen Asphyxiation?

A) It is unconsciousness caused by Nitrogen can lead to death

B) It is unconsciousness caused by Inert gas can lead to death

C) Nitrogen, which is non-flammable, non-toxic and colourless

D) Nitrogen which is undetectable to the human senses

Correct Answer: A (It is unconsciousness caused by Nitrogen can lead to death)

Explanation: Nitrogen asphyxiation occurs when nitrogen displaces oxygen in an enclosed space, leading to an oxygen-deficient atmosphere that can cause unconsciousness or death due to hypoxia.



- A) Toxicity is the degree to which a substance or mixture of substances can harm humans.
- B) Toxicity means poisonous and affects you on oral consumption
- C) Toxicity means Acidic
- D) None of the above

Correct Answer: A (Toxicity is the degree to which a substance or mixture of substances can harm humans.)

Explanation: Toxicity is a measure of the inherent ability of a substance to cause harmful effects in humans, whether through inhalation, ingestion, or skin absorption, regardless of its flammability.

Q5. What is Time Weighted Average?

- A) Exposure time
- B) Short Term exposure time
- C) Highest exposure time
- D) Averaging all the exposure times for a time period 8 hours

Correct Answer: D (Averaging all the exposure times for a time period 8 hours)

Explanation: Time Weighted Average (TWA) represents the average concentration of a substance to which a person may be safely exposed over a specified time period, typically 8 hours, without suffering long-term health effects.

Q6. Which is a non-accumulator of charges?

- A) Toluene is a Non-accumulator of charges
- B) Cyclohexane is a Non-accumulator of charges
- C) Alcohols is an Accumulator of charges
- D) Ketones is a Non-accumulator of charges

Correct Answer: D (Ketones is a Non-accumulator of charges)

Explanation: Ketones are classified as non-accumulators of static charge because they possess sufficient conductivity to dissipate charge build-up during cargo transfer. Toluene and cyclohexane are hydrocarbons with low conductivity (accumulators), while alcohols are generally polar but often treated differently; ketones are the standard example of a non-accumulator in this context.

Q7. What is Auto-ignition?

- A) The ignition of a combustible material without initiation by a spark or flame, when the material has been raised to a temperature at which self-sustaining combustion occurs.
- B) Automatic ignition of material in a hot atmosphere
- C) Ignition of materials when they come in contact with each other
- D) Ignition caused by mixture of vapours given off by cargoes

Correct Answer: A (The ignition of a combustible material without initiation by a spark or flame, when the material has reached a temperature at which self-sustaining combustion occurs.)

Explanation: Auto-ignition refers to the property of a substance whereby it ignites spontaneously without the requirement of an external ignition source like a spark or flame. This occurs once the material reaches its auto-ignition temperature in the presence of an oxidant.

Q8. Rapid combustion takes place in coal mines.

- A) True
- B) False

Correct Answer: A (True)

Explanation: The accumulation of coal dust and trapped gases in mines can lead to rapid oxidation and combustion. When these reach critical concentrations, they can cause rapid, often explosive, combustion events.



- A) cooking gas
- B) nitrogen gas
- C) oxygen gas
- D) producer gas

Correct Answer: C (oxygen gas)

Explanation: Oxygen is the primary supporter of combustion in the fire triangle. Without a sufficient concentration of oxygen, the chemical reaction of fire cannot be sustained.

Q10. Sun's heat and lightning strike cause spontaneous combustion in forests.

- A) True
- B) False

Correct Answer: A (True)

Explanation: Spontaneous combustion in forests, often referred to as wildfires, can be triggered by natural phenomena such as lightning strikes or extreme solar heat drying out vegetation until it reaches its ignition point.

Q11. Substances which catch fire are called acids.

- A) True
- B) False

Correct Answer: B (False)

Explanation: Substances that catch fire easily are termed 'flammable' or 'combustible' substances. Acids are a chemical classification based on pH levels and are not synonymous with flammable materials.

Q12. Burning of candle is an example of slow oxidation.

- A) True
- B) False

Correct Answer: B (False)

Explanation: Burning is a form of rapid oxidation accompanied by heat and light. Slow oxidation refers to processes like rusting of iron or respiration, not the combustion of a candle.

Q13. Tank clingage depends on:

- A) surface tension only
- B) cohesive and adhesive forces
- C) cohesive, adhesive forces and surface tension
- D) cohesive forces, adhesive forces and the liquids melting point

Correct Answer: C (cohesive, adhesive forces and surface tension)

Explanation: Tank clingage is the amount of cargo remaining on tank surfaces after discharge, influenced by the complex interaction of the liquid's surface tension, and the adhesive/cohesive forces between the liquid and the tank bulkhead.

Q14. Oxygen Analyzer: what is your key finding in measuring oxygen level?

- A) Portable oxygen analyzers are normally used to determine whether the atmosphere inside an enclosed space (cargo tank for example) may be considered fully inerted or safe for entry.
- B) Fixed oxygen analyzers can be used for monitoring the oxygen content of the boiler uptakes and the inert gas main.
- C) The common types of oxygen analyzers in use are, Paramagnetic sensors and Electrochemical sensors.
- D) Oxygen is strongly paramagnetic whereas most other common gases are not.



Explanation: The primary objective of using a portable oxygen analyzer on a tanker is to ensure the atmosphere is safe for human entry (21% O₂) or to verify the efficacy of inert gas systems to prevent combustion.

Q15. Key Features and Measurement of an Effective Safety Culture; identify one statement that identifies an effective safety culture

- A) Recognition that all accidents are preventable and only usually occur following unsafe actions or a failure to follow established procedures
- B) Individual seafarers assume responsibility for safety rather than relying on others to provide it
- C) Through mutual respect, increasing confidence in the value of the safety culture results in a more effective Safety Management System
- D) A safety management system that is always setting targets for continuous improvement, with a goal of zero accidents

Correct Answer: D (A safety management system that is always setting targets for continuous improvement, with a goal of zero accidents)

Explanation: An effective safety culture is characterized by the continuous pursuit of excellence. Setting measurable targets for improvement and striving for the goal of zero accidents is a cornerstone of modern safety management systems.

Q16. What is the maximum interval of time between sampling and analysing for each sampling head location sequentially?

- A) Not exceeding 50 minutes interval
- B) Not exceeding 40 minutes interval
- C) Not exceeding 30 minutes interval
- D) Not exceeding 60 minutes interval

Correct Answer: C (Not exceeding 30 minutes interval)

Explanation: In fixed gas sampling systems, the regulation requires that the sequence of sampling from various heads is frequent enough to ensure safety. An interval not exceeding 30 minutes is the industry standard to ensure real-time hazard detection.

Q17. For entering an enclosed space, 21% _____ is necessary.

- A) Oxygen
- B) carbondioxide
- C) Nitrogen

Correct Answer: A (Oxygen)

Explanation: Normal atmospheric air contains approximately 21% oxygen. This concentration is the prerequisite for safe human entry into enclosed spaces to prevent hypoxia.

Q18. What is the least a person should do before entering machinery spaces?

- A) Wear appropriate personal protective equipment
- B) Inform the bridge
- C) Inform his superiors

Correct Answer: A (Wear appropriate personal protective equipment)

Explanation: Personnel must always prioritize personal safety. Wearing the correct Personal Protective Equipment (PPE) is the fundamental requirement before entering any machinery space to mitigate known hazards.



- A) Solid cellulosic materials such as wood, paper, clothing, etc.
- B) Vapour/air mixture over the surface of flammable liquids
- C) Energised electrical equipment
- D) Combustible materials such as magnesium, sodium, etc.

Correct Answer: D (Combustible materials such as magnesium, sodium, etc.)

Explanation: Class D fires are specifically defined as those involving combustible metals, such as magnesium, titanium, sodium, or potassium, which require specialized extinguishing agents.

Q20. The hydrocarbon content in the atmosphere of a pump room is measured using a

- A) Tankscope
- B) Explosimeter
- C) The Draeger instrument
- D) Alarm system detector

Correct Answer: C (The Draeger instrument)

Explanation: Draeger-type gas detection tubes are commonly used on ships to provide accurate, localized concentration readings of specific hydrocarbon gases in various compartments, including pump rooms.

Q21. What is the limitation of a combustible gas detector of catalytic combustion type?

- A) It cannot be used for measurement in air
- B) It cannot be used for measurement in nitrogen
- C) It cannot be used for measurement in methane
- D) It cannot be used for measurement in oxygen

Correct Answer: B (It cannot be used for measurement in nitrogen)

Explanation: Catalytic combustion (pellistor) sensors require oxygen to oxidize the hydrocarbon sample on the filament. Therefore, they cannot function in an inerted environment (e.g., nitrogen) where oxygen is absent.

Q22. The manifold pressure when loading oil cargo should be

- A) Above the cargo tank pressure
- B) Below the cargo tank pressure
- C) Above atmospheric pressure
- D) 'Zero'

Correct Answer: D ('Zero')

Explanation: During cargo operations, manifold pressure is maintained at a level that balances the system; maintaining 'zero' pressure relative to the tank prevents excessive stress on hoses and potential leaks.

Q23. What is the maximum reliable shelf life of a filter gas mask canister if the seal is unbroken?

- A) 1 year from the date of manufacture
- B) 3 years from the date of manufacture
- C) 5 years from the date of manufacture
- D) 7 years from the date of manufacture

Correct Answer: C (5 years from the date of manufacture)

Explanation: The shelf life of filter canisters is limited by the degradation of the absorbent materials over time. Most manufacturers stipulate a 5-year maximum shelf life for sealed canisters to guarantee integrity.



- A) middle
- B) top
- C) bottom

Correct Answer: B (top)

Explanation: Gas vapours with a relative density lower than air (lighter than air) naturally accumulate in the highest parts of an enclosed space due to buoyancy.

Q25. Which step is NOT generally taken when gas-freeing a tank?

- A) Washing the tank interior with sea water
- B) Application of degreasing solvents
- C) Removal of corrosion products and sludge
- D) Fresh air ventilation

Correct Answer: B (Application of degreasing solvents)

Explanation: Gas-freeing involves ventilation and physical cleaning to remove hazards. The application of degreasing solvents is a specific cleaning process that is not a standard, mandatory step for routine gas-freeing.

Q26. The reading of a combustible gas indicator indicates the percentage of the

- A) lower explosive limit of a flammable gas concentration
- B) upper explosive limit of a flammable gas concentration
- C) concentration of flammable gas in a compartment
- D) concentration by weight of nonflammable gas in a compartment

Correct Answer: A (lower explosive limit of a flammable gas concentration)

Explanation: A combustible gas indicator typically provides readings as a percentage of the Lower Explosive Limit (LEL), indicating how close the current gas concentration is to forming an explosive mixture.

Q27. A crew member suffering from hypothermia should be given _____ .

- A) small dose of alcohol
- B) treatment for shock
- C) a large meal
- D) a brisk rub down

Correct Answer: B (treatment for shock)

Explanation: A person with hypothermia requires gentle, gradual warming and stabilization. Alcohol is strictly forbidden as it can lead to vasodilation and further heat loss, and massage (rubbing) is dangerous as it moves cold blood to the heart.

Q28. What instrument is used to measure hydrocarbon gas in an inert atmospheres?

- A) Explosimeter
- B) The Draeger instrument
- C) The Multi-Gas detector
- D) Tankscope

Correct Answer: D (Tankscope)

Explanation: A Tankscope is specifically designed to measure hydrocarbon gas concentrations in inerted atmospheres where oxygen levels are low and standard explosimeters would fail.



- A) The infra-red (IR) sensor is a transducer for the measurement of the concentration of hydrocarbons by the absorption of infra-red radiation.
- B) The vapor to be monitored reaches the measuring chamber by diffusion or by means of a pump.
- C) An infrared source illuminates a volume of gas that has entered inside the measurement chamber.
- D) The gas absorbs some of the infrared wavelengths as the light passes through it, while others pass through it completely unattenuated.

Correct Answer: D (The gas absorbs some of the infrared wavelengths as the light passes through it, while others pass through it completely unattenuated.)

Explanation: Infrared (IR) sensors work by measuring the absorption of specific infrared light wavelengths as they pass through the gas. Hydrocarbons absorb specific IR frequencies, allowing the concentration to be calculated.

Q30. Areas like cargo residues, cargo and tank coatings must be suspected for oxygen-deficient areas.

- A) True
- B) False

Correct Answer: A (True)

Explanation: Cargo residues, sludge, and even tank coatings can oxidize or undergo biological processes that consume oxygen, creating localized or overall oxygen-deficient atmospheres in tanks.

Q31. What should be prohibited in the vicinity of battery rooms?

- A) Painting
- B) Washing
- C) Smoking

Correct Answer: C (Smoking)

Explanation: Battery rooms generate hydrogen gas during charging, which is highly flammable. Smoking is strictly prohibited to prevent the ignition of potentially explosive hydrogen-air mixtures.

Q32. Safe ship operations include:

- A) Running the ship profitably
- B) Deliver cargo safely
- C) No damage to the environment

Correct Answer: C (No damage to the environment)

Explanation: Safe ship operations are defined by protecting life, property, and the marine environment. Preventing environmental damage is a core component of modern international maritime safety standards.

Q33. Which of the following current level leads to muscular contraction?

- A) 0.01A
- B) 0.015A
- C) 0.5A
- D) 0.10A

Correct Answer: B (0.015A)

Explanation: Electrical current as low as 10-15 milliamperes (0.01-0.015A) can cause involuntary muscular contractions, potentially preventing a person from letting go of an energized source.



- A) The Master will decide based on findings by Chief Officer
- B) The chief officer will assess the situation and decide
- C) Yes, the checklist, permit and other procedures apply for all enclosed spaces
- D) No, the pumproom has a fixed fan for ventilation and frequently visited

Correct Answer: C (Yes, the checklist, permit and other procedures apply for all enclosed spaces)

Explanation: Under the IMO and industry standards, all pump rooms are considered enclosed spaces. Entry requires strict adherence to permits, testing, and checklist protocols regardless of their ventilation status.

Q35. What should you do when the alarm bell on a self-contained breathing apparatus sounds?

- A) Immediately evacuate the contaminated area.
- B) Open the bypass valve on the regulator and immediately evacuate the contaminated area.
- C) Move the tank selector lever to the full tank position and reset the alarm so you can evacuate the area when it sounds again.
- D) Move the reserve lever to the 'reserve' position on the regulator and reset the alarm so you can evacuate the area when it sounds again.

Correct Answer: A (Immediately evacuate the contaminated area.)

Explanation: The alarm on a self-contained breathing apparatus (SCBA) signifies the low-pressure warning, indicating the air reserve is nearing depletion. The only safe action is to immediately retreat to a safe, non-contaminated environment.

Q36. Experience indicates that hazardous potentials in respect of static electricity do not occur if the velocity of liquid, while loading is: Find the correct answer

- A) Below 7~m/s
- B) Above 7~m/s
- C) Above 10~m/s
- D) Below 3~m/s

Correct Answer: A (Below 7~m/s)

Explanation: To minimize the risk of static electricity generation caused by the flow of liquid through pipelines, flow velocities must be strictly controlled, typically staying below 7 m/s during initial loading stages.

Q37. Electrochemical Sensors: identify a key finding for its use on board tankers

- A) Electrochemical sensors are based on the fact that cells can be constructed that react with the measured gas and generate an electric current.
- B) The Cell current can be measured, and the amount of gas determined.
- C) The sensors are low cost and are small enough to allow several to be incorporated into the same instrument, making them suitable for use in multi-gas detectors.
- D) There are numerous electrochemical sensors available covering a number of gases which may be present in the shipboard environment.

Correct Answer: C (The sensors are low cost and are small enough to allow several to be incorporated into the same instrument)

Explanation: Electrochemical sensors are compact, energy-efficient, and can be easily miniaturized, which makes them ideal for inclusion in portable multi-gas detectors used by seafarers for daily safety checks.

Q38. Which of the following conditions represents a particular advantage of using a positive pressure type self-contained breathing apparatus in an atmosphere that is immediately dangerous to life or health?



- A) The equipment is compact and the wearer can work in confined spaces without difficulty.
- B) The equipment used is lightweight and easy to wear by reducing physical strain on the wearer.
- C) The average operating time for most air cylinders is over an hour.
- D) The positive pressure in the face piece prevents contaminated air from entering the face piece.

Correct Answer: D (The positive pressure in the face piece prevents contaminated air from entering the face piece.)

Explanation: Positive pressure SCBAs maintain a slightly higher pressure inside the mask than the surrounding environment. This design ensures that if there is a minor seal failure, air leaks out instead of allowing toxic air in.

Q39. Which of the following is an entry activity?

- A) Permit closed
- B) Confirmation of removal of all equipment
- C) Entry permit issued

Correct Answer: C (Entry permit issued)

Explanation: An entry permit is a formal document issued prior to entering an enclosed space, establishing the safety conditions, testing requirements, and authorizations necessary for a safe entry.

Q40. Which of the following is a particular challenge when entering fuel oil tanks?

- A) slippery surface
- B) corrosion
- C) low lighting
- D) sediments

Correct Answer: A (slippery surface)

Explanation: Fuel oil tanks contain viscous residues that create extremely slippery internal surfaces, posing a significant risk of falls for personnel performing inspections or maintenance.

Q41. Water extinguishes a fire by

- A) Heat removal or cooling
- B) Smothering or oxygen exclusion
- C) Flame inhibition

Correct Answer: A (Heat removal or cooling)

Explanation: Water extinguishes fires primarily by absorbing heat (cooling effect). This reduces the fuel temperature below its ignition point, thereby breaking the fire triangle.

Q42. What is an EEBD used for?

- A) For escape from machinery or accommodation spaces in the event of a fire
- B) For safety when entering a tank
- C) For rescuing a person from a tank
- D) For supplying oxygen to a victim

Correct Answer: A (For escape from machinery or accommodation spaces in the event of a fire)

Explanation: An EEBD (Emergency Escape Breathing Device) is specifically designed to provide a limited supply of air for short-duration escape from an accommodation or machinery space during a fire or emergency.



- A) flame scanner
- B) halide torch
- C) combustible gas indicator
- D) detector filament

Correct Answer: C (combustible gas indicator)

Explanation: A combustible gas indicator is the standard instrument used to detect the presence and concentration of flammable vapours, protecting against explosive atmospheres.

Q44. A man has suffered a burn on the arm. There is extensive damage to the skin with charring present. How is this injury classified using standard medical terminology?

- A) Dermal burn
- B) Third-degree burn
- C) Major burn
- D) Lethal burn

Correct Answer: B (Third-degree burn)

Explanation: A third-degree burn involves full-thickness damage to the skin, including the dermis and subcutaneous layers, often resulting in charring or white, leathery tissue.

Q45. Which of the following is required to be included in the fireman's (emergency) outfit?

- A) Chemical protection face shield
- B) Approved work vest
- C) Self-contained breathing apparatus
- D) 5 cell approved flashlight

Correct Answer: C (Self-contained breathing apparatus)

Explanation: A Self-Contained Breathing Apparatus (SCBA) is a mandatory component of the fireman's outfit as per SOLAS requirements, enabling entry into smoke-filled or oxygen-deficient spaces.

Q46. What is the least you must do before entering a cofferdam in the engine room?

- A) Open up the manhole and enter the cofferdam
- B) Complete an enclosed space checklist
- C) Enter the cofferdam and simultaneously start ventilation

Correct Answer: B (Complete an enclosed space checklist)

Explanation: Entering any enclosed space, including a cofferdam, requires a formal permit and verified safety checklist to ensure the atmosphere is safe and all hazards are controlled.

Q47. Which of the following activity is done on completion of work?

- A) Closing and securing all openings
- B) Satisfactory lighting
- C) Intimating all the related parties about the intended entry

Correct Answer: A (Closing and securing all openings)

Explanation: Upon completion of work, it is a critical safety practice to secure all manholes and openings to prevent unauthorized entry and ensure the integrity of the compartment.



- A) Heat removal or cooling
- B) Smothering or oxygen exclusion
- C) Flame inhibition

Correct Answer: B (Smothering or oxygen exclusion)

Explanation: Carbon dioxide is a non-flammable gas that is heavier than air. It functions as an extinguishing agent by displacing oxygen from the area, effectively smothering the fire.

Q49. When the percentage oxygen is between _____ one may feel, fainting, nausea vomiting.

- A) 19.5% and 16%
- B) 16% and 12%
- C) 12% and 10%
- D) 10% and 8%

Correct Answer: D (10% and 8%)

Explanation: Oxygen levels between 8% and 10% are extremely dangerous, often leading to rapid unconsciousness, severe neurological impairment, and failure of critical bodily functions.

Q50. If there has been a fire in a closed unventilated compartment it may be unsafe to enter because of

- A) excess hydrogen
- B) unburned carbon particles
- C) excess nitrogen
- D) a lack of oxygen

Correct Answer: D (a lack of oxygen)

Explanation: Fires consume oxygen rapidly, and the combustion products often include carbon monoxide and other toxic gases. An unventilated space post-fire is almost certainly oxygen-deficient.

Practice Set 3

Q1. Which are the international bodies that provide regulatory standards for electrical equipment certification?

- A) IEEC and CENELEC
- B) IEC and CENELEC
- C) IEEC and ISO
- D) IMO and ISO

Correct Answer: B (IEC and CENELEC)

Explanation: The IEC (International Electrotechnical Commission) and CENELEC (European Committee for Electrotechnical Standardization) are the primary bodies that set global standards for the certification of electrical equipment used in hazardous locations.

Q2. Why is the 'maximum surface temperature' identified, for electrical equipment used in hazardous areas onboard LNG carriers?

- A) The limit is on minimum surface temperature on LNG carriers as the cargo is cryogenic
- B) Equipment should operate within these limits to reduce the power and boil-off consumption
- C) Equipment should operate within these limits to prevent the formation of a hot surface
- D) Equipment should operate within these limits to reduce the heat transfer to the cryogenic cargo



Correct Answer: C (Equipment should operate within these limits to prevent the formation of a hot surface)

Explanation: In hazardous areas, limiting the maximum surface temperature of equipment ensures that it does not reach the auto-ignition temperature of any flammable gases or vapours that might be present.

Q3. How is intrinsic safety achieved for electrical circuits used in hazardous areas?

- A) By limiting the ignition energy and surface temperature that can arise during operation
- B) By limiting the number of equipment used in a space
- C) By increasing mechanical ventilation to the area
- D) By installing machinery which have higher power and thus shorter operation time

Correct Answer: A (By limiting the ignition energy and surface temperature that can arise during operation)

Explanation: Intrinsic safety design limits the electrical energy (voltage and current) within a circuit to a level that is too low to produce the sparks or heat required to ignite an explosive atmosphere.

Q4. Where can electrical equipment be classified as Ex-rated equipment?

- A) In Zones 0, 1 and 2
- B) In Zones 1 and 2
- C) In Zones 0 and 2
- D) In Zones 0 and 1

Correct Answer: B (In Zones 1 and 2)

Explanation: Electrical equipment used in hazardous areas must be rated for the specific zone classification. Ex-rated equipment is generally utilized in Zones 1 (where explosive gas is likely) and 2 (where it is unlikely but possible).

Q5. Pick the IEC symbol for explosion proof electrical protection?

- A) ex-e
- B) ex-n
- C) ex-d
- D) ex-p

Correct Answer: C (ex-d)

Explanation: The symbol 'ex-d' denotes 'flameproof' (or explosion-proof) equipment, where the enclosure is designed to withstand an internal explosion without igniting the surrounding atmosphere.

Q6. In cleaning up an oil spill, straw is an example of a

- A) chemical agent
- B) blocker
- C) sorbent

Correct Answer: C (sorbent)

Explanation: Straw is an organic material with natural absorbent properties used to soak up spilled oil. In maritime pollution response, such materials are classified as sorbents because they collect liquid through absorption or adsorption.

Q7. Heavy fuel oils when spilled are _____.

- A) more harmful to sea life than lighter oils
- B) easier to clean up than lighter refined oils
- C) less harmful to sea life than lighter oils
- D) not a real threat to marine life

Correct Answer: C (less harmful to sea life than lighter oils)

Explanation: Heavy fuel oils are more persistent and tend to form thick mats or sink, causing physical smothering of marine



Q8. For a vessel approaching a port of reception, what the vessel must check before the transfer of residues from the ship?

- A) Ports and terminals must provide reception facilities
- B) Ports and terminals must have Certificates of Adequacy
- C) Ports must Apply for a Certificate of Adequacy.
- D) Port Reception facility operations procedures

Correct Answer: B (Ports and terminals must have Certificates of Adequacy)

Explanation: Under MARPOL regulations, the Port State must ensure that port reception facilities are adequate. Masters must verify that the terminal possesses a valid Certificate of Adequacy (CoA) to ensure the legal and compliant disposal of ship-generated residues.

Q9. Which one of the following discharges have the most long-term ecological impact due to pollution?

- A) Water ballast in oil fuel tanks.
- B) Ballast added to cargo tanks.
- C) Discharges of clean and segregated ballast
- D) Machinery space bilges.

Correct Answer: A (Water ballast in oil fuel tanks.)

Explanation: Ballast water in oil fuel tanks becomes heavily contaminated with oil residues, which are persistent and hazardous if discharged. Unlike clean ballast, this mixture has significant long-term ecological consequences due to its toxicity and environmental persistence.

Q10. What assigns for each new ship a calculable figure that will denote its total emissions of CO2 from combustion of fuel?

- A) Energy Efficiency Design Index
- B) Energy Inefficiency Design Index
- C) SOX Design Index
- D) VOX Design Index

Correct Answer: A (Energy Efficiency Design Index)

Explanation: The Energy Efficiency Design Index (EEDI) is a mandatory technical measure under MARPOL Annex VI. It provides a specific value for new ships to quantify their carbon emissions relative to their capacity and performance, encouraging energy-efficient ship design.

Q11. When did Annex VI take effect?

- A) May 19, 2005
- B) October 12, 1991
- C) February 25, 1997
- D) July 3, 2015

Correct Answer: A (May 19, 2005)

Explanation: MARPOL Annex VI, which addresses air pollution from ships, including sulfur oxides (SOx) and nitrogen oxides (NOx) emissions, officially entered into force on May 19, 2005.



- A) Vessel type, size and speed
- B) Vessel flag, type and age
- C) Vessel type, size and crew
- D) Vessel type, size and age

Correct Answer: D (Vessel type, size and age)

Explanation: MARPOL requirements are stringent and evolve based on the vessel's specific type (e.g., oil tanker vs. general cargo), its size (deadweight/tonnage), and its age, as newer vessels must meet stricter construction and equipment standards.

Q13. MARPOL applies to:

- A) applies to all ships, oil platforms, barges and private yachts
- B) tankers greater than 150GT and cargo ships greater than 400GT
- C) applies to all ships, oil platforms, barges and private yachts and hydrofoils
- D) applies to all ships, oil platforms, barges and private yachts and hovercrafts

Correct Answer: A (applies to all ships, oil platforms, barges and private yachts)

Explanation: MARPOL is a comprehensive convention that applies to all ships, including commercial vessels, private yachts, barges, and offshore platforms, to prevent pollution from operational and accidental causes.

Q14. When you have completed bunkering operations, the hoses should be _____ .

- A) blown down with inert gas
- B) drained into drip pans or tanks
- C) stowed with their ends open for venting
- D) steam cleaned and flushed with hot water

Correct Answer: B (drained into drip pans or tanks)

Explanation: After bunkering, hoses must be drained to prevent spillage from residual oil remaining in the line. Draining into drip trays or designated slop tanks ensures that the residual oil is contained and managed correctly.

Q15. What is the most important regulatory requirement for the master arriving at the lightering position?

- A) Lightering of Oil and Hazardous Material Cargoes Pre-arrival notices.
- B) Reporting of incidents.
- C) Designation of lightering zones.
- D) Factors considered in designating lightering zones.

Correct Answer: A (Lightering of Oil and Hazardous Material Cargoes Pre-arrival notices.)

Explanation: Lightering involves transferring cargo between ships at sea. The Master must adhere to strict regulatory requirements regarding pre-arrival notices to ensure the operation is authorized, safe, and complies with environmental pollution prevention standards.

Q16. Most minor spills of oil products are caused by

- A) equipment failure
- B) human error
- C) major casualties
- D) unforeseeable circumstances

Correct Answer: B (human error)

Explanation: Statistically, the majority of oil spills during routine shipboard operations occur due to human error, such as improper valve alignment, failing to monitor levels, or poor communication during liquid transfers.



- A) Don required PPE as determined from the MSDS.
- B) Fight fire (if any), being careful to use firefighting methods compatible with the material involved.
- C) Shut off or otherwise stem the spill at the source.
- D) Contain liquid material using barriers, such as absorbents, rags or other equipment suitable to dam the flow.

Correct Answer: A (Don required PPE as determined from the MSDS.)

Explanation: Personnel safety is the paramount priority in any emergency response. According to the Material Safety Data Sheet (MSDS), wearing the appropriate personal protective equipment (PPE) is the first step to prevent exposure to hazardous substances before attempting containment.

Q18. Segregated ballast tanks are required to be fitted on

- A) Every crude oil and new product tanker of 20,000dwt and above
- B) Every crude oil tanker of 20,000dwt and above and every new product tanker of 30,000dwt and above
- C) Every crude oil and new product tanker of 30,000dwt and above

Correct Answer: B (Every crude oil tanker of 20,000dwt and above and every new product tanker of 30,000dwt and above)

Explanation: MARPOL Annex I requires that crude oil tankers of 20,000 dwt and above, and new product tankers of 30,000 dwt and above, be fitted with Segregated Ballast Tanks (SBT) to prevent oil-water mixing.

Q19. What is used to incinerate solid and liquid wastes arising from the operation of the ship?

- A) Incinerator
- B) Oily water separation
- C) ISM Screen
- D) Fire Sprinkler Cleaner

Correct Answer: A (Incinerator)

Explanation: An incinerator is the designated shipboard equipment used for the controlled burning of solid and liquid wastes, as permitted under MARPOL Annex V and Annex VI, to reduce waste volume and manage shipboard garbage.

Q20. The Volatile Organic Compound (cargo vapor) Management Plan is required on what type of vessel?

- A) Tank sensors
- B) Container vessels
- C) Ro-ro vessels
- D) Crude Oil Carriers

Correct Answer: D (Crude Oil Carriers)

Explanation: The Volatile Organic Compound (VOC) Management Plan is mandatory for crude oil carriers, as these vessels handle cargoes that release significant amounts of VOCs, which must be managed to comply with environmental regulations.

Q21. Your vessel is carrying 24,000 barrels (3816 cubic metres) of oil for discharge. The cargo hoses have an inside diameter of eight inches. The container around the loading manifold must hold

- A) three barrels
- B) four barrels
- C) six barrels
- D) eight barrels

Correct Answer: A (three barrels)

Explanation: Under industry standards, the containment capacity of the drip tray beneath the cargo manifold is sized based on the



Q22. It is generally NOT allowed to clean up an oil spill by using _____ .

- A) boom
- B) suction equipment
- C) chemical agents
- D) skimmers

Correct Answer: C (chemical agents)

Explanation: Chemical agents, such as dispersants, are strictly regulated and generally not allowed for oil spill cleanup without specific approval, as they can cause deeper ecological damage or toxicity to the marine environment compared to mechanical recovery methods like booms and skimmers.

Q23. The total quantity of oil discharged at sea on new tankers must not exceed:

- A) 1/15000 of the total quantity of that particular cargo of which the residue formed part
- B) 1/2000 of the total quantity of that particular cargo of which the residue formed part
- C) 1/25000 of the total quantity of that particular cargo of which the residue formed part
- D) 1/30000 of the total quantity of that particular cargo of which the residue formed part

Correct Answer: D (1/30000 of the total quantity of that particular cargo of which the residue formed part)

Explanation: MARPOL regulations strictly limit the discharge of oil residues. For a new tanker, the total quantity of oil which may be discharged into the sea is restricted to no more than 1/30,000 of the total quantity of the particular cargo which the residue formed part.

Q24. A starting interlock in the ODME ensures

- A) the monitoring equipment is fully operational before commencing the discharge of the oil effluent at sea
- B) prevents malfunctioning of the system
- C) that an alarm is given off when the system malfunctions
- D) stops discharge of the oil effluent when the oil content exceeds 5 part per million

Correct Answer: A (the monitoring equipment is fully operational before commencing the discharge of the oil effluent)

Explanation: The starting interlock on an Oil Discharge Monitoring Equipment (ODME) system ensures that the system is fully calibrated and functional before any discharge begins, preventing accidental illegal discharge of oily water.

Q25. A gas venting system on a tanker that has gas lines in each tank connected to a common gas main running the full length of the ship is called:

- A) gas-main system
- B) A common-main system
- C) A vent-main system
- D) An independent system

Correct Answer: B (A common-main system)

Explanation: In a common-main system, multiple cargo tanks are connected to a single, shared gas main running the length of the ship, allowing for efficient venting or inerting of the cargo tank atmosphere.

Q26. The inert gas is delivered to the cargo tanks by an inert gas deck main line. Find the correct labelling of the deck equipment items.



- A) 1-Pressure/Vacuum Valve; 2- Cargo tank valves; 3- Mast Riser; 4-Isolating Valve
- B) 1-Mast Riser; 2-Isolating Valve; 3-Cargo tank valves; 4- Pressure/Vacuum Valve
- C) 1-Cargo tank valves; 2- Pressure/Vacuum Valve; 3- Mast Riser; 4-Isolating Valve
- D) 1-Isolating Valve; 2- Pressure/Vacuum Valve; 3- Mast Riser; 4-Cargo tank valves

Correct Answer: C (1-Cargo tank valves; 2- Pressure/Vacuum Valve; 3- Mast Riser; 4-Isolating Valve)

Explanation: The proper sequence for deck equipment in the IG distribution line is the cargo tank isolating valve, then the PV valve, followed by the mast riser and final isolating valves, ensuring pressure control and safety of the tank boundary.

Q27. LNG liquid pressure is the function of liquid density and liquid level.

- A) True
- B) False

Correct Answer: A (True)

Explanation: Hydrostatic pressure at any point in a tank is directly proportional to the density of the liquid and the vertical height (head) of the liquid column above that point.

Q28. Pneumatically Operating Valve; what is the most critical component for correct valve operation, based on your shipboard experience of running such a system?

- A) Valve orifice
- B) Valve stem
- C) The Diaphragm
- D) The valve springs

Correct Answer: C (The Diaphragm)

Explanation: The diaphragm in a pneumatically operated valve is the most critical component, as it acts as the sealing/actuating mechanism; if it fails or ruptures, the valve loses its ability to respond to control signals or maintain its position.

Q29. Advantages of using inert gas, pick one that is the main reason for its application,

- A) No explosive mixtures can form in the tank.
- B) Reduces corrosion.
- C) Voyage cleaning of tanks is unnecessary.
- D) Reduces pumping time because of positive pressure in the tanks.

Correct Answer: A (No explosive mixtures can form in the tank.)

Explanation: The primary objective of an Inert Gas System is to maintain the oxygen content within the cargo tanks below the level required for combustion, thereby preventing the formation of an explosive atmosphere.

Q30. Which statement about a centrifugal cargo pump is TRUE?

- A) It is a positive displacement pump.
- B) It must have a positive suction.
- C) Increasing rotation speed will decrease discharge pressure.

Correct Answer: B (It must have a positive suction.)

Explanation: Centrifugal pumps are not self-priming and require a positive suction head to operate, as they rely on the fluid filling the impeller casing to create the centrifugal force necessary to move liquid.

Q31. The total capacity of the slop tank or tanks should be not less than what percentage of the oil-carrying capacity of the vessel?



- A) 3%
- B) 5%
- C) 7%
- D) 10%

Correct Answer: A (3%)

Explanation: MARPOL regulations specify that the slop tank capacity must be sufficient to hold the oily water generated during tank cleaning and cargo residue, which is set at a minimum of 3% of the ship's total oil-carrying capacity.

Q32. Acidic solutions are formed in cargo tanks of tankers

- A) when sulphur compounds in oil cargoes react with water
- B) when sulphur compounds in oil cargoes react with oxygen
- C) when sulphur compounds in oil cargoes react with hydrocarbon gas
- D) when sulphur compounds in oil cargoes react with chemicals

Correct Answer: A (when sulphur compounds in oil cargoes react with water)

Explanation: When sulfur compounds inherent in crude oil react with moisture or water introduced into the cargo tanks, they form acidic solutions (like sulfuric acid), which significantly increase the rate of corrosion of the tank structure.

Q33. Tank level measurement of free water by gauging tape and plumb bob: find the labeling combination that is correct,

- A) 1 Water Innage; 2-Water Cut; 3- Water Cut; 4- Water Ullage
- B) 1- Water Cut; 2- Water Innage; 3- Water Cut; 4- Water Ullage
- C) 1- Water Cut; 2- Water Innage; 3- Water Ullage; 4- Water Cut
- D) 1- Water Cut; 2- Water Cut; 3- Water Innage; 4- Water Ullage

Correct Answer: B (1- Water Cut; 2- Water Innage; 3- Water Cut; 4- Water Ullage)

Explanation: Gauging terminology distinguishes between 'innage' (the depth of liquid) and 'cut' (the depth of a specific component like water at the bottom). Correct labeling of water cut and innage is essential for accurate cargo volume determination.

Q34. An inert gas generator burns what oil to generate inert gas?

- A) Crude oil
- B) Heavy oil
- C) Diesel oil
- D) Gas oil

Correct Answer: D (Gas oil)

Explanation: Inert gas generators are designed to burn high-quality fuel, typically marine gas oil, to produce a clean, soot-free, and high-quality inert gas suitable for pumping into cargo tanks.

Q35. Float gauges can only record liquid level greater than _____ inches.

- A) Four
- B) five
- C) ten

Correct Answer: A (Four)

Explanation: Float gauges operate using a buoyant element that rests on the surface of the liquid. Due to the mechanical design and the physical dimensions of the float, they have a minimum measurable level, typically around four inches.



- A) An isolating valve
- B) The deck water seal
- C) A valve provided in the scrubber
- D) The deck mechanical non-return valve

Correct Answer: B (The deck water seal)

Explanation: The deck water seal is a mandatory safety device in the inert gas system that acts as a non-return barrier, preventing hazardous cargo gases from flowing back into the inert gas plant and engine room.

Q37. Continuous exposure of the float to the fluctuation in sea level can damage the _____ .

- A) Tape tensioning device
- B) diaphragm
- C) sensors

Correct Answer: A (Tape tensioning device)

Explanation: The tape tensioning device is vital for maintaining the accuracy of the gauge. Continuous, harsh exposure to changing sea levels and movement causes mechanical wear on the tensioning spring and tape, potentially leading to inaccurate readings.

Q38. The float gauge is provided with a _____ valve.

- A) Ball
- B) gate
- C) pressure

Correct Answer: B (gate)

Explanation: Float gauges are fitted with a gate valve to allow for maintenance and isolation of the instrument from the tank atmosphere, ensuring that the gauge can be serviced without breaching the integrity of the tank.

Q39. Tank coating requirements developed by IMO are applicable to

- A) new crude oil tankers of 5000 GRT and above
- B) new crude oil tankers of 5000 DWT and above
- C) new crude oil tankers of 500 GRT and above
- D) new crude oil tankers of 500 DWT and above

Correct Answer: B (new crude oil tankers of 5000 DWT and above)

Explanation: IMO performance standards for protective coatings (PSPC) require that new crude oil tankers of 5,000 DWT and above are fitted with robust coating systems to prevent corrosion and structural degradation.

Q40. A deepwell pump is a type of _____ .

- A) screw pump
- B) centrifugal pump
- C) eductor
- D) gear pump

Correct Answer: B (centrifugal pump)

Explanation: A deepwell pump is a type of centrifugal pump mounted directly into the tank. This design eliminates long pipelines and allows for efficient stripping of cargo, commonly found on chemical and product tankers.



- A) is done to remove traces of the previous cargo
- B) will remove heavy wax sediments at the tank bottom
- C) is not effective when carrying a small quantity of refined products
- D) cannot be done with chemical solvents

Correct Answer: A (is done to remove traces of the previous cargo)

Explanation: Bottom washing is a specific tank cleaning procedure focused on the lowest part of the tank to remove sediment and traces of the previous cargo, particularly important when switching to a different cargo grade.

Q42. Float gauge is a commonly used device in most of the tankers.

- A) True
- B) False

Correct Answer: A (True)

Explanation: Float gauges are a standard and reliable technology used for measuring liquid levels in cargo tanks, providing a direct mechanical or remote readout of the cargo ullage.

Q43. An inert gas system is required to be fitted:

- A) On all tankers of above 20,000 tonnes deadweight
- B) On product tankers of above 20,000 tonnes deadweight
- C) On crude oil tankers of above 20,000 tonnes deadweight
- D) On chemical tankers of above 20,000 tonnes deadweight

Correct Answer: C (On crude oil tankers of above 20,000 tonnes deadweight)

Explanation: Inert Gas Systems (IGS) are mandatory for crude oil tankers of 20,000 tonnes deadweight and above to ensure that cargo tank atmospheres remain non-flammable throughout the voyage and cargo operations.

Q44. Radar gauge is a type of tank gauging equipment.

- A) True
- B) False

Correct Answer: A (True)

Explanation: Radar gauging is a modern, accurate, and non-contact method of measuring liquid levels in tanks, which is widely accepted and used in the maritime industry for cargo management.

Q45. Centrifugal pumps are otherwise known as

- A) Reciprocating pumps
- B) Gravity pumps
- C) Positive displacement pumps
- D) Hydro-dynamic pumps

Correct Answer: D (Hydro-dynamic pumps)

Explanation: Centrifugal pumps are classified as hydro-dynamic pumps because they impart velocity energy to the fluid using a rotating impeller, which is then converted into pressure energy.



- A) VLCCs
- B) Crude oil tankers
- C) Medium range tankers
- D) Product carriers

Correct Answer: D (Product carriers)

Explanation: The ring main pipeline system is common on product carriers because it allows for multiple, simultaneous cargo segregations, enabling the vessel to load or discharge different products without risk of cross-contamination.

Q47. Class B fires are caused by

- A) combustible solids
- B) flammable gases
- C) flammable liquids
- D) electricity

Correct Answer: C (flammable liquids)

Explanation: Class B fires involve flammable liquids (such as fuel oil, diesel, or lubricants). Effective firefighting for these fires typically requires foam, CO2, or dry chemical powder to smother the flames.

Q48. Use of Pressure-Vacuum (PV) Relief Valve; label the diagram correctly:

- A) 1-IG Main; 2-Deck Isolating Valve; 3- Vent Main; 4- By-Pass Valve; 5- PV valve; 6- Tank Isolating Valve
- B) 1-Deck Isolating Valve; 2- Vent Main; 3- IG Main; 4- By-Pass Valve; 5- PV valve; 6- Tank Isolating Valve
- C) 1-Deck Isolating Valve; 2- IG Main; 3- Vent Main; 4- By-Pass Valve; 5- PV valve; 6- Tank Isolating Valve
- D) 1-Deck Isolating Valve; 2- IG Main; 3- Vent Main; 4- By-Pass Valve; 5- Tank Isolating Valve; 6- PV valve

Correct Answer: C (1-Deck Isolating Valve; 2- IG Main; 3- Vent Main; 4- By-Pass Valve; 5- PV valve; 6- Tank Isolating Valve)

Explanation: The Pressure/Vacuum (PV) valve setup includes isolating valves and a bypass to allow for testing or maintenance. Option C provides the correct sequence for managing the connection between the inert gas main, vent line, and the tank.

Q49. Cargo tank fill alarm indicator panels:

- A) are normally fitted on the bridge and sometimes on deck
- B) are normally fitted in the engine control room and sometimes on deck
- C) are normally fitted on the bridge and sometimes in the engine control room
- D) are normally fitted in the cargo control room and sometimes on deck

Correct Answer: D (are normally fitted in the cargo control room and sometimes on deck)

Explanation: Cargo tank high-level and overfill alarms are critical safety instruments; they are typically monitored in the Cargo Control Room (CCR) where the transfer is being managed, with local alarms often provided on deck.

Q50. Identify the equipment in the IG system correctly:

- A) 1-O2 Analyzer; 2-Demister; 3-Air Intake-Gas Freeing; 4- Regulating Valves; 5- Blowers
- B) 1- Demister; 2-O2 Analyzer; 3-Air Intake-Gas Freeing; 4- Regulating Valves; 5- Blowers
- C) 1- Demister; 2-O2 Analyzer; 3 - Blowers; 4- Regulating Valves; 5- Air Intake-Gas Freeing
- D) 1- Demister; 2-O2 Analyzer; 3-Regulating Valves; 4-Air Intake-Gas Freeing; 5- Blowers

Correct Answer: B (1- Demister; 2-O2 Analyzer; 3-Air Intake-Gas Freeing; 4- Regulating Valves; 5- Blowers)

Explanation: The standard IG system flow path includes the scrubber (demister), blowers, O2 analyzer, regulating valves for flow control, and air intake for gas freeing, arranged to ensure clean and safe delivery of inert gas.



Practice Set 4

Q1. Working instructions specifying the manner in which the ship is to be loaded and ballasted to avoid unacceptable stresses in the vessel's structure is required for tankers:

- A) Greater than 50 metres in length
- B) Greater than 75 meters in length
- C) Greater than 100 meters in length
- D) Greater than 150 meters in length **Correct answer : Greater than 150 meters in length**

Correct Answer: D (Greater than 150 meters in length **Correct answer : Greater than 150 meters in length)**

Explanation: Regulation regarding structural integrity and loading calculations for tankers requires that ships over 150 meters in length carry specialized loading instruments and documentation to monitor shear forces and bending moments.

Q2. A self-contained breathing apparatus is used to

- A) make underwater repairs to barges
- B) determine if the air in a tank is safe for men
- C) enter areas that may contain dangerous fumes or lack oxygen
- D) resuscitate an unconscious person **Correct answer : enter areas that may contain dangerous fumes or lack oxygen**

Correct Answer: C (enter areas that may contain dangerous fumes or lack oxygen)

Explanation: Self-contained breathing apparatus (SCBA) is essential personal protective equipment (PPE) for entering hazardous environments where the atmosphere may be oxygen-deficient or contaminated with toxic cargo vapours.

Q3. Identify correct labelling for Tanker VEC

- A) 1- P/V Valve; 2- P/V Breaker; 3- Mast Riser; 4 - Deck water seal
- B) 1- P/V Valve; 2- P/V Breaker; 3- Deck water seal; 4- Mast Riser
- C) 1- Mast Riser; 2- P/V Breaker; 3- P/V Valve; - Deck water seal
- D) 1- P/V Valve; 2- Mast Riser; 3- P/V Breaker; 4 - Deck water seal

Correct Answer: D (1- P/V Valve; 2- Mast Riser; 3- P/V Breaker; 4 - Deck water seal)

Explanation: In a standard tanker VEC (Vapour Emission Control) system configuration, the sequence involves the P/V valve (pressure regulation), Mast Riser (venting), P/V Breaker (vacuum protection), and the deck water seal as the primary flame arrestor.

Q4. A combustible gas indicator will operate correctly ONLY when the

- A) hydrocarbon content of the atmosphere is less than the U.E.L.
- B) atmosphere is deficient in oxygen
- C) compartment to be tested is free of CO₂ **Correct answer : hydrocarbon content of the atmosphere is less than the U.E.L.**

Correct Answer: A (hydrocarbon content of the atmosphere is less than the U.E.L.)

Explanation: Combustible gas indicators are designed to measure gas concentrations up to the Upper Explosive Limit (U.E.L.). Above this level, the mixture is too rich to burn, potentially resulting in inaccurate or misleading readings.



- A) Transfer of bulk liquids, physical checks
- B) Transfer of bulk liquids - elements to be verified verbally
- C) Additional considerations for Bulk Liquid Chemicals
- D) Additional considerations for Bulk Liquefied Gases **Correct answer : Transfer of bulk liquids, physical checks**

Correct Answer: A (Transfer of bulk liquids, physical checks)

Explanation: Part A of the ISGOTT Ship/Shore Safety Check-list is designated for physical verification of safety measures. It ensures that critical items, such as hose connections and bonding, are checked manually before cargo transfer commences.

Q6. The approval period for a shipboard Oil Pollution Emergency Plan expires after

- A) two years
- B) three years
- C) four years
- D) five years **Correct answer : five years**

Correct Answer: D (five years **Correct answer : five years)**

Explanation: The Shipboard Oil Pollution Emergency Plan (SOPEP) is a mandatory requirement under MARPOL Annex I, Regulation 37. It must be reviewed and updated regularly, with approval typically valid for a five-year cycle to align with the ship's safety management system and periodic surveys.

Q7. Which of the listed functions is the purpose of a gas scrubber in an inert gas generation system?

- A) Cools the inert gas.
- B) Maintains the oxygen content at 5% by volume.
- C) Bleeds off static electricity in the inert gas.
- D) Maintains flow to the water seal on the gas main. **Correct answer : Cools the inert gas.**

Correct Answer: A (Cools the inert gas.)

Explanation: In an Inert Gas System (IGS), the gas scrubber is essential for cooling the hot exhaust gases from the boiler or inert gas generator using water. It also performs the critical function of removing sulfur oxides and soot particles from the flue gas.

Q8. With an increase in temperature the volume of flammable and combustible liquids

- A) expands
- B) contracts
- C) remains constant
- D) remains constant if pressure remains constant **Correct answer : expands**

Correct Answer: A (expands)

Explanation: Flammable and combustible liquids possess a coefficient of thermal expansion, meaning their volume increases as temperature rises. This physical property necessitates the implementation of ullage spaces in cargo tanks to prevent over-pressurization due to thermal expansion.

Q9. The standard unit of liquid volume used in the petroleum industry, as well as the tanker industry, is a

- A) barrel
- B) drum
- C) gallon
- D) liter **Correct answer : barrel**

Correct Answer: A (barrel)

Explanation: In the global petroleum and tanker industry, the standard unit of measurement for volume is the barrel. One barrel is



Q10. Combustible gas indicators operate by drawing an air sample into the instrument

- A) over an electrically heated platinum filament
- B) where it is mixed with nitrogen
- C) where it is ignited by a sparking device
- D) where its specific gravity is measured **Correct answer : over an electrically heated platinum filament**

Correct Answer: A (over an electrically heated platinum filament)

Explanation: Combustible gas indicators (explosimeters) utilize a Wheatstone bridge circuit. An air sample is drawn over an electrically heated platinum filament; the combustion of flammable gas on the filament increases its temperature and resistance, which is then measured and displayed as a percentage of the Lower Explosive Limit (LEL).

Q11. Each ship having an inert gas system must have a portable instrument to measure concentrations of hydrocarbon vapor in inert atmospheres and also to measure

- A) nitrogen
- B) oxygen
- C) carbon dioxide
- D) water vapor **Correct answer : oxygen**

Correct Answer: B (oxygen)

Explanation: Ships with an inert gas system are required to maintain portable monitoring equipment to verify the atmosphere within cargo tanks. This must include an oxygen analyzer to ensure the oxygen content is maintained below the critical limit (typically 5% or 8% by volume) to prevent combustion.

Q12. As required by Reg.31 of Annex 1 of Marpol the ODMCS should be provided with a recording device for recording the discharge in litres per nautical mile and total quantity discharged. This discharge should be kept onboard at least for a period of

- A) Intermittent, 1 year
- B) Continuous, 1 year
- C) Intermittent, 3 years
- D) Continuous, 3 years **Correct answer : Continuous, 3 years**

Correct Answer: D (Continuous, 3 years **Correct answer : Continuous, 3 years)**

Explanation: MARPOL Annex I, Regulation 31 requires the Oil Discharge Monitoring and Control System (ODMCS) to have a recording device to track discharges. These records must be preserved for at least three years to be available for inspection by Port State Control or Flag State authorities.

Q13. Which is the most accurate instrument for measuring the amount of oxygen in the atmosphere of a confined space?

- A) Combustible gas indicator
- B) Oxygen indicator
- C) Flame safety lamp **Correct answer : Oxygen indicator**

Correct Answer: B (Oxygen indicator)

Explanation: An oxygen indicator is the most accurate device for measuring oxygen concentration in confined spaces, providing a direct percentage reading. Flame safety lamps are obsolete for this purpose, and combustible gas indicators are specifically designed for detecting flammable vapors, not oxygen levels.



- A) CO2
- B) fresh air
- C) the tested compartment's air
- D) water **Correct answer : fresh air**

Correct Answer: B (fresh air)

Explanation: To ensure accuracy and prevent sensor damage or 'memory effects' from previous samples, the oxygen indicator must be purged with fresh ambient air. This restores the sensor to the baseline atmospheric oxygen concentration (approx 20.9%) before the next measurement.

Q15. The maneuvering vessel normally berths on which side during STS operations?

- A) Any side to
- B) Port side to
- C) Starboard side to
- D) As instructed by the mother vessel **Correct answer : Starboard side to**

Correct Answer: C (Starboard side to)

Explanation: During Ship-to-Ship (STS) transfer operations, the maneuvering vessel (the ship coming alongside) typically berths on the starboard side of the mother vessel. This allows the master on the maneuvering ship to use the standard bridge layout, where the conning position favors starboard-side maneuvering.

Q16. When the cofferdam to be purged to check the leakage and remove the leaked oil?

- A) Before starting the cargo operation
- B) Before starting the cargo operation and after the cargo operation is completed
- C) During the cargo operation
- D) After the cargo operation is completed **Correct answer : Before starting the cargo operation and after the cargo operation is completed**

Correct Answer: B (Before starting the cargo operation and after the cargo operation is completed)

Explanation: Cofferdams adjacent to cargo tanks must be purged and checked before cargo operations commence to ensure no leakage is present, and again after completion to verify the integrity of the bulkheads during the operation. This is a vital practice for preventing pollution and fire hazards.

Q17. What does "I" stand for in the centrifugal pump diagram?

- A) Eye (Suction)
- B) Shaft
- C) Impeller
- D) Diffuser **Correct answer : Shaft**

Correct Answer: B (Shaft)

Explanation: In standard centrifugal pump diagrams, the central rotating component supported by the bearing housing that transmits power from the motor to the impeller is referred to as the shaft.

Q18. The area indicated by the number "III" in the pump diagram is known as the

- A) Speed Torque Control Valve
- B) Cofferdam Purging
- C) Local Control Valve
- D) Cargo Purging Connection **Correct answer : Speed Torque Control Valve**

Correct Answer: A (Speed Torque Control Valve)



Q19. How the Framo pump has almost eliminated cavitation for free flowing liquids?

- A) Volute casing is designed in such a way, that its smoothness avoids cavitation
- B) The backwards curved vanes eliminate the cavitation
- C) The hydrofoil design of the curved vanes, eliminate cavitation
- D) Since the suction pipe is eliminated, cavitation is also eliminated for liquids with free flow **Correct answer : Since the suction pipe is eliminated, cavitation is also eliminated for liquids with free flow**

Correct Answer: D (Since the suction pipe is eliminated, cavitation is also eliminated for liquids with free flow **Correct answer : Since the suction pipe is eliminated, cavitation is also eliminated for liquids with free flow)**

Explanation: Framo submerged cargo pumps eliminate the suction pipe by being placed directly at the bottom of the cargo tank. By removing the long, tortuous suction piping, the pressure drop is minimized, effectively preventing cavitation for free-flowing liquids.

Q20. State if the following is True or False? After fixing the snap on hydraulic couplings of portable Framo, secure it with locking ring-only then the arrangement is safe.

- A) True
- B) False **Correct answer : True**

Correct Answer: A (True)

Explanation: For portable hydraulic pumps, the snap-on couplings are high-pressure connections. Securing the locking ring is mandatory to prevent accidental disconnection due to vibration or pressure surges, which could cause high-pressure hydraulic oil spray and pump failure.

Q21. Select the correct statement regarding pressure.

- A) Pressure type vacuum valves can prevent Backflow.
- B) Dynamic pressure is a function of the fluid velocity and its density.
- C) The water reservoir of a self-priming pump may be at the back of the impeller.
- D) Stripping pumps are not of the self-priming type. **Correct answer : Dynamic pressure is a function of the fluid velocity and its density.**

Correct Answer: B (Dynamic pressure is a function of the fluid velocity and its density.)

Explanation: Bernoulli's principle establishes that dynamic pressure is directly related to the density of the fluid and the square of its velocity. This relationship is fundamental to understanding pump head, flow velocity, and pressure energy conversion within the pump casing.

Q22. How portable pumps are used?

- A) Potable pumps are used in parallel with the main Framo pump to discharge the tank in a short time
- B) When main pump is not working, portable pumps are used to discharge the liquid cargo to the terminal
- C) Potable pumps are used in series with the main Framo pump to discharge the tank with large pressure.
- D) When main pump is not working, the portable pump is used to transfer the cargo from this tank to another empty tank, where the main pump is working **Correct answer : When main pump is not working, the portable pump is used to transfer the cargo from this tank to another empty tank, where the main pump is working**

Correct Answer: D (When main pump is not working, the portable pump is used to transfer the cargo from this tank to another empty tank, where the main pump is working)

Explanation: Portable pumps are primarily used as emergency or backup equipment. If a fixed main cargo pump fails, a portable pump can be lowered into the tank to facilitate cargo transfer to another tank where a functioning main pump can then discharge it to the terminal.



- A) Eye (Suction)
- B) Shaft
- C) Impeller
- D) Diffuser **Correct answer : Eye (Suction)**

Correct Answer: A (Eye (Suction))

Explanation: The eye of a centrifugal pump is the central inlet point where the fluid enters the pump impeller. It is the area of lowest pressure, and its design is crucial for preventing cavitation.

Q24. Which number represents the Hydraulic Motor in the pump cross-section illustration?

- A) I
- B) II
- C) III
- D) IV **Correct answer : I**

Correct Answer: A (I)

Explanation: In a cross-section illustration of a submerged cargo pump, 'I' typically marks the hydraulic motor section. This motor is powered by the vessel's hydraulic system and drives the impeller directly.

Q25. The part "IX" in the centrifugal pump diagram indicates

- A) Stuffing Box
- B) Shaft Sleeve
- C) Packing
- D) Casing **Correct answer : Packing**

Correct Answer: C (Packing)

Explanation: In a pump assembly diagram, 'IX' identifies the packing or mechanical seal arrangement. These components are critical for preventing cargo leakage from the pump into the hydraulic motor section or the engine room.

Q26. Why a two way valve is provided in the air or inert gas line provided to check the leakage of oil in the cofferdam?

- A) One is for the normally closed position and the other is for checking the leakage of hydraulic oil in the motor side
- B) One is for the normally closed position and the other is for checking the leakage of cargo oil in the pump side
- C) One is for the normally closed position, the other is for checking the leakage of cargo oil in the pump side and the third one is for checking the leakage of hydraulic oil in the motor side **Correct answer : One is for the normally closed position, the other is for checking the leakage of cargo oil in the pump side and the third one is for checking the leakage of hydraulic oil in the motor side**

Correct Answer: C (One is for the normally closed position, the other is for checking the leakage of cargo oil in the pump side and the third one is for checking the leakage of hydraulic oil in the motor side)

Explanation: The cofferdam purging/monitoring system uses a multi-way valve arrangement to ensure that the cavity between the cargo pump and hydraulic motor can be isolated. This allows for independent testing for both cargo oil leakage and hydraulic oil leakage using inert gas or air.



- A) Hydraulic Motor
- B) Volute Casing
- C) Impeller
- D) Anti-rotation Brake **Correct answer : Volute Casing**

Correct Answer: B (Volute Casing)

Explanation: In a typical cargo pump diagram, the volute casing refers to the spiral-shaped housing surrounding the impeller. It is designed to convert the kinetic energy of the fluid (velocity) into pressure energy as it moves toward the discharge nozzle.

Q28. How vacuum drain is done in Framo pumps?

- A) Pump has a different vacuum line which is connected to the cargo tank well; vacuum pump is used to suck the oil out to make the well dry
- B) Vacuum is created in the stripping line to pump the cargo out to make the well dry
- C) Vacuum drain is used to check leakage in pump's cofferdam
- D) vacuum drain is part of the pump casing **Correct answer : Pump has a different vacuum line which is connected to the cargo tank well; vacuum pump is used to suck the oil out to make the well dry**

Correct Answer: A (Pump has a different vacuum line which is connected to the cargo tank well; vacuum pump is used to suck the oil out to make the well dry)

Explanation: Framo cargo pumps utilize a specialized vacuum drain line connected to the tank well. By applying a vacuum, the system can clear the remaining cargo from the well even when the main pump has lost suction due to low levels.

Q29. The number "10" in the Tanker cargo pump assembly indicates a _____.

- A) Top Plate
- B) Stripping Valve
- C) Resilient Mounting **Correct answer : Stripping Valve**

Correct Answer: B (Stripping Valve)

Explanation: The assembly number '10' in typical Framo pump diagrams corresponds to the stripping valve. This valve is essential for final tank drainage, allowing for the evacuation of residual cargo after the main pumping phase.

Q30. Why the cargo pump in the Framo cannot be overloaded?

- A) The pressure of hydraulic oil to the motor driving the cargo pump is designed in such a way that, the flow is always kept constant
- B) The pressure of hydraulic oil to the motor driving the cargo pump is designed in such a way that, the pump runs at the optimum speed
- C) The pressure of hydraulic oil to the motor driving the cargo pump is designed in such a way that, when the pump over speeds, the flow of oil is reduced
- D) The pressure of hydraulic oil to the motor driving the cargo pump is designed in such a way that, the pressure can run the pump at 75% of the capacity **Correct answer : The pressure of hydraulic oil to the motor driving the cargo pump is designed in such a way that, the pump runs at the optimum speed**

Correct Answer: B (The pressure of hydraulic oil to the motor driving the cargo pump is designed in such a way that, the pump runs at the optimum speed)

Explanation: Framo pumps are designed with an integrated hydraulic control system that matches the power delivery to the load. By maintaining optimum speed and preventing excessive power draw, the pump system prevents mechanical overloading of the components.



- A) Vane
- B) Eye of impeller
- C) Discharge nozzle
- D) None of the above **Correct answer : Discharge nozzle**

Correct Answer: C (Discharge nozzle)

Explanation: In a centrifugal pump, the fluid is accelerated by the impeller and forced into the discharge nozzle. This component guides the fluid out of the pump casing into the main cargo piping system.

Q32. Normally how much hydraulic oil is required to run the Framo portable pump?

- A) 360 litres per min
- B) 250 litres per min
- C) 140 litres per min
- D) 90 litres per min **Correct answer : 140 litres per min**

Correct Answer: C (140 litres per min)

Explanation: Framo portable pumps are specifically rated for hydraulic flow requirements to achieve their design performance. For standard portable pumps, this requirement is typically 140 liters per minute of hydraulic oil.

Q33. Select the correct statement regarding operation of the portable pumps?

- A) Before using portable pump, rig the pump on the deck and while pumping, lower the hose in the tank
- B) When the liquid cargo level goes down, lower the hose accordingly, keeping the rigged pump on the deck
- C) Make sure that the portable pump is completely submerged and lowered till the bottom of the tank and then start the pumping operation
- D) Ensure that only the lower half of the pump is submerged and is lowered as the cargo level falls. **Correct answer : Ensure that only the lower half of the pump is submerged and is lowered as the cargo level falls.**

Correct Answer: D (Ensure that only the lower half of the pump is submerged and is lowered as the cargo level falls.)

Explanation: During operation, the portable pump must be lowered as the liquid level drops to maintain suction. Submerging only the lower half ensures the intake remains flooded while preventing the pump from resting on bottom sediments that could clog it.

Q34. What is the pressure of the hydraulic oil in the return side after running the hydraulic motor?

- A) Around 3 Bar
- B) Around 10 bar
- C) Around 15 bar
- D) Above 15 bar **Correct answer : Around 3 Bar**

Correct Answer: A (Around 3 Bar)

Explanation: In a closed-loop hydraulic system, the return pressure must be kept low to ensure efficient operation of the hydraulic motor. A pressure of approximately 3 bar is standard for the return side of these systems.

Q35. What does "V" indicate in the pump illustration?

- A) Speed Torque Control Valve
- B) Cofferdam Purging
- C) Local Control Valve
- D) Cofferdam Check Pipe **Correct answer : Cofferdam Purging**

Correct Answer: B (Cofferdam Purging)

Explanation: In the context of Framo submerged pump monitoring, 'V' refers to the purging port or cofferdam purging connection. This is used to test for seal integrity by applying air/gas pressure and checking for leaks.



- A) Hydraulic Motor
- B) Sleeve
- C) Wear rings
- D) Mechanical Oil Seal **Correct answer : Mechanical Oil Seal**

Correct Answer: D (Mechanical Oil Seal **Correct answer : Mechanical Oil Seal)**

Explanation: The component 'VII' represents the mechanical oil seal. This seal is a critical barrier separating the cargo area from the hydraulic oil area, preventing contamination between the two.

Q37. Normally what is the discharge capacity of Framo portable pumps?

- A) 150 cubic meter per hour
- B) 100 cubic meter per hour
- C) 70 cubic meter per hour
- D) 200 cubic meter per hour **Correct answer : 70 cubic meter per hour**

Correct Answer: C (70 cubic meter per hour)

Explanation: Portable centrifugal pumps used for tanker stripping or cargo transfer are typically sized for specific capacities. A 70 cubic meter per hour capacity is a common standard for these portable hydraulic-driven units.

Q38. What is the function of the item marked "II" in the pump diagram?

- A) Impeller
- B) Hydraulic Motor
- C) Anti-rotation Brake
- D) Suction Well **Correct answer : Anti-rotation Brake**

Correct Answer: C (Anti-rotation Brake)

Explanation: The anti-rotation brake (or stay) is essential for submerged pumps to ensure the pump body remains stable and does not rotate under the reaction forces generated by the spinning impeller.

Q39. In a centrifugal pump:

- A) The discharge side has a negative pressure.
- B) The fluid moves in a direction opposite to the impeller.
- C) The fluid moves in the same direction as the impeller.
- D) The width of the discharge pipeline decreases outwards. **Correct answer : The fluid moves in the same direction as the impeller.**

Correct Answer: C (The fluid moves in the same direction as the impeller.)

Explanation: In a centrifugal pump, the fluid enters the impeller eye and is thrown outwards by centrifugal force. The impeller vanes guide the fluid in the same rotational direction until it enters the volute casing.

Q40. What does "V" indicate in the below illustration?

- A) Bearings
- B) Mechanical Oil Seal
- C) Suction Well
- D) Sleeve **Correct answer : Suction Well**

Correct Answer: C (Suction Well)

Explanation: The 'suction well' (or inlet screen/well area) is the area 'V' where the cargo enters the pump. It ensures that the fluid is presented to the impeller eye as smoothly as possible.



- A) Impeller
- B) Discharge Nozzle
- C) Eye of Impeller
- D) None of the above **Correct answer : Eye of Impeller**

Correct Answer: C (Eye of Impeller)

Explanation: The 'eye' of the impeller (labeled 'VIII') is the central inlet where the liquid enters the impeller vanes, representing the transition point from the suction pipe/well to the kinetic energy section.

Q42. How purging and seal monitoring is done?

- A) Air or inert gas pipe connected to the line, the two way valve which remains closed is moved to hydraulic oil side and air or inert gas is opened and if there is any leak, it is purged out and the quantity is measured.
- B) Air or inert gas pipe connected to the line, the two way valve is moved to hydraulic oil side and air or inert gas is opened and if there is any leak, it is purged out and the quantity is measured, then the two way valve is moved to the cargo oil side and the air or inert gas is opened and if there is any leak, it is purged out and the quantity is measured
- C) Air or inert gas pipe connected to the line, the two way valve is moved to cargo oil side and air or inert gas is opened and if there is any leak, it is purged out and the quantity is measured. **Correct answer : Air or inert gas pipe connected to the line, the two way valve is moved to hydraulic oil side and air or inert gas is opened and if there is any leak, it is purged out and the quantity is measured, then the two way valve is moved to the cargo oil side and the air or inert gas is opened and if there is any leak, it is purged out and the quantity is measured**

Correct Answer: B (Air or inert gas pipe connected to the line, the two way valve is moved to hydraulic oil side and air or inert gas is opened and if there is any leak, it is purged out and the quantity is measured, then the two way valve is moved to the cargo oil side and the air or inert gas is opened and if there is any leak, it is purged out and the quantity is measured)

Explanation: Seal monitoring involves a two-step purge process. First, checking the hydraulic side for oil leakage, then switching the valve to check the cargo side. This ensures both seals are intact and that neither cargo nor hydraulic oil is leaking into the cofferdam.

Q43. Bernoulli's principle states that:

- A) pressure (P) x height (H) is a constant
- B) height (H) x velocity (V) is a constant
- C) pressure (P) x velocity (V) is a constant
- D) pressure (P) x density (D) is a constant **Correct answer : pressure (P) x velocity (V) is a constant**

Correct Answer: C (pressure (P) x velocity (V) is a constant)

Explanation: Bernoulli's principle, in its simplified form for fluid dynamics in pumps, expresses the conservation of energy, where increases in velocity occur alongside decreases in static pressure, and vice versa.

Q44. From the below illustration, identify the name of the marking "14".

- A) Resilient Mounting
- B) Deck Trunk
- C) Cargo Pipe
- D) Hydraulic Pressure Pipe **Correct answer : Cargo Pipe**

Correct Answer: C (Cargo Pipe)

Explanation: In the assembly schematic, '14' represents the cargo discharge pipe. This pipe conveys the pressurized fluid from the pump outlet up through the deck trunk to the vessel's cargo manifold.



- A) Diffuser
- B) Shaft
- C) Eye (Suction)
- D) Impeller **Correct answer : Diffuser**

Correct Answer: A (Diffuser)

Explanation: The diffuser 'V' is a stationary component in the pump casing that guides the fluid from the impeller outlet. Its primary function is to efficiently convert the high-velocity kinetic energy of the fluid into static pressure.

Q46. How the speed of the portable pump can be varied?

- A) Portable pump speed cannot be varied
- B) Using the flow control valve of the main cargo pump
- C) By varying the speed of the feed pump
- D) The portable pump is always rigged with its own flow control valve which is used for speed variation **Correct answer : The portable pump is always rigged with its own flow control valve which is used for speed variation**

Correct Answer: D (The portable pump is always rigged with its own flow control valve which is used for speed variation)

Explanation: The speed of a hydraulic motor is proportional to the flow rate of the hydraulic oil. Therefore, the pump is rigged with a flow control valve that regulates the oil volume, allowing for precise speed variation.

Q47. What kind of pumps can move extremely thick fluids like sludge without clogging?

- A) Rotary vane pumps
- B) Reciprocating pumps
- C) Deep well pumps
- D) Screw pumps **Correct answer : Screw pumps**

Correct Answer: D (Screw pumps **Correct answer : Screw pumps)**

Explanation: Screw pumps are positive displacement pumps known for their ability to handle high-viscosity fluids and sludges efficiently. Their design prevents clogging and ensures a steady flow regardless of the fluid thickness.

Q48. What is the pressure of the hydraulic oil in the inlet side for running the hydraulic motor?

- A) Around 50 Bar
- B) Around 170 bar
- C) Around 100 bar
- D) Around 25 bar **Correct answer : Around 170 bar**

Correct Answer: B (Around 170 bar)

Explanation: Submerged hydraulic cargo pumps are designed to operate at high pressure to achieve the necessary power for liquid discharge. An inlet pressure of approximately 170 bar is typical for these motors under full-load conditions.

Q49. The reciprocating pump creates a suction effect when there is a

- A) decrease in pressure
- B) increase in pressure
- C) no change in pressure
- D) decrease or an increase in pressure **Correct answer : decrease in pressure**

Correct Answer: A (decrease in pressure)

Explanation: A reciprocating pump operates by the motion of a piston or plunger. When the piston moves to expand the volume of the pump chamber, it creates a decrease in pressure (a vacuum), which draws the fluid in through the suction valve.



- A) Eye of impeller
- B) Shaft
- C) Impeller
- D) Casing **Correct answer : Impeller**

Correct Answer: C (Impeller)

Explanation: The impeller 'VI' is the rotating core of the centrifugal pump. It imparts kinetic energy to the liquid by spinning at high speed, driving the fluid from the eye towards the periphery of the casing.

Practice Set 5

Q1. Select the correct point regarding the operation of the portable pumps?

- A) Before using portable pump, rig the pump on the deck and while pumping, lower the hose in the tank
- B) When the liquid cargo level goes down, lower the hose accordingly, keeping the rigged pump on the deck
- C) Make sure that the portable pump is completely submerged and lowered till the bottom of the tank and then start the pumping operation
- D) Ensure that only the lower half of the pump is submerged and is lowered as the cargo level falls.

Correct Answer: D (Ensure that only the lower half of the pump is submerged and is lowered as the cargo level falls.)

Explanation: When operating portable cargo pumps (such as submersible hydraulic or pneumatic pumps) on a tanker, it is critical to ensure the pump's suction end/lower half remains submerged in the liquid cargo, lowering it progressively as the level falls. This prevents dry running, loss of prime, and air entrainment while avoiding unnecessary full submersion of the pump's power head.